UNIT – IV : TESTING AND PERFORMANCE OF I.C. ENGINES

4.1 Why test an engine ?

- To determine **power, efficiency, fuel consumption** & heat distribution; to compare engines, to check design & after repair; for emission certification; to draw performance curves.

4.2 Parameters of Performance

- **Indicated power (IP)** : power developed in the cylinder (from indicator diagram).

$$IP = p_{im} L A n N_c / 60000 \text{ (kW)} \quad [p \text{ in } N/m^2, L \text{ in } m, A \text{ in } m^2]$$

- n = number of cylinders ; **N_c = N/2** for 4-stroke (power strokes / min), = N for 2-stroke ; N = rpm.
- **Brake power (BP)** : useful power at the crankshaft, measured by a dynamometer.

$$BP = 2\pi NT / 60000 \text{ (kW)}, \; T \text{ in } N\cdot m$$

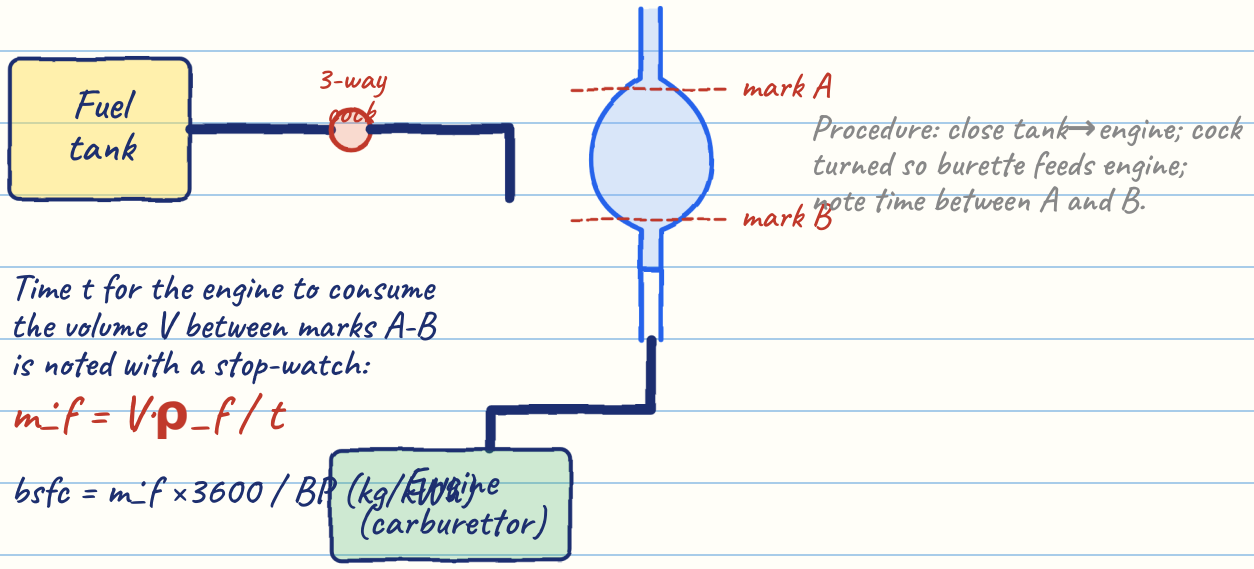
- **Friction power (FP)** = IP – BP (bearing, piston-ring, valve gear, pumping, accessories).
- **Mechanical efficiency** η_m = BP / IP = BP/(BP + FP) (≈ 80-90 %).
- **Mean effective pressure** : the hypothetical constant pressure acting on the piston for one stroke giving the same work : imep (from IP), **bmep** = BP × 60000 / (L A n N_c), **fmep** = imep – bmep.
- **Indicated thermal efficiency** η_{ith} = IP / (m_f × CV); **Brake thermal efficiency** η_{bth} = BP / (m_f × CV) ; m_f in kg/s, CV in kJ/kg.
- **Specific fuel consumption** : **bsfc** = m_f(kg/h) / BP (kW) kg/kWh; isfc similarly with IP. ηbth = 3600 / (bsfc × CV).

- **Volumetric efficiency** η_v = actual air drawn / swept volume at atm. conditions (80-90 % SI, 85-95 % CI).
- **Relative (efficiency ratio) efficiency** = η_{th}(actual) / $\eta_{air\text{-}standard}$.
- **Air-fuel ratio**, **specific output** (kW/l of swept volume), **mean piston speed**, **torque**.

4.3 Measurement of Cylinder Pressure (Indicators)

- Objective : p-V / p-θ diagram ⇒ imep, IP. **Mechanical indicators** (Watt / Crosby / Dobbie-McInnes) work only at low speed because of inertia.
- **Electronic indicators** : the pressure transducer in the cylinder head (flush-mounted) + charge amplifier + crank-angle encoder + oscilloscope / DAQ.
- Types : **Piezo-electric** (quartz - best; high natural frequency > 100 kHz), strain gauge, capacitive, optical, magnetic (inductance) pick-up.
- The p-θ diagram is converted to p-V using the slider-crank relation; **TDC detection** is critical (error of 1° gives ~ 3-5 % error in imep).

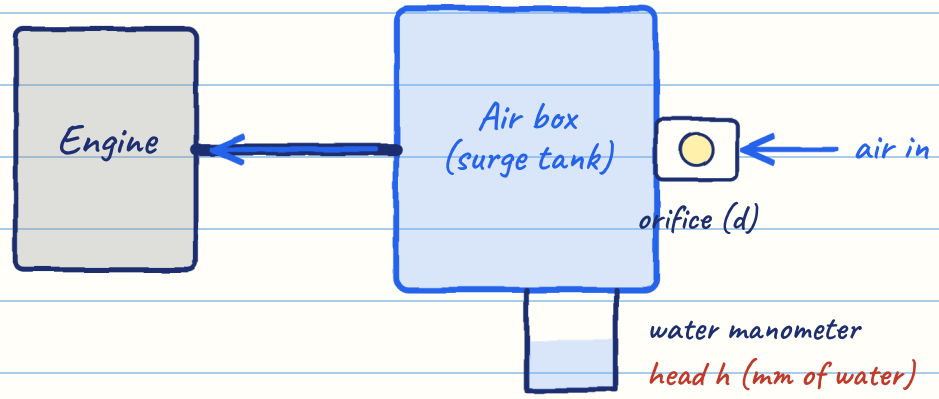
4.4 Measurement of Fuel Consumption



- (a) **Volumetric method** : burette with 2 marks (25, 50, 100 cc) ; time t for known volume V is noted ⇒ $\dot{m}_f = V\rho / t$. Simple, accurate for small engines.
- (b) **Gravimetric method** : fuel tank on a weighing balance, time for a fixed mass (e.g. 50 g) consumed. Independent of temperature (density).
- (c) **Flow meters** : positive displacement (PD), Coriolis, turbine; continuous reading (for large engines & ECU testing).
- For **diesel**, injector return line must be returned downstream of the measurement.

4.5 Measurement of Air Intake

Air-box method of air-flow measurement



Volume flow $V = C_d \cdot (\pi/4)\, d^2 \cdot \sqrt{(2\, g\, h_a)}\ (m^3/s)$, $h_a = h \cdot \rho_w / \rho_a$

Box damps pulsations of the engine suction; box volume ≈ 500 × swept volume.

- (a) Air-box with orifice (above) : pulsation damped; $V = Cd\ (\pi/4) d^2 \sqrt{(2\, g\, h_a)}$; typical $Cd = 0.6$; $m_a = \rho_a V$.
- (b) Viscous-flow (Alcock) meter : laminar flow through a honeycomb element ⇒ $\Delta p \propto$ flow (linear) ⇒ no damping tank required.
- (c) Hot-wire (MAF) sensors, turbine / vane meters, thermal mass flow - used in vehicles & modern tests.
- (d) Pitot / nozzle & indirect : from the A/F ratio using exhaust analysis (carbon balance method).

4.6 Measurement of Exhaust Gas Composition

- Purpose : combustion quality, A/F ratio, emissions (CO, HC, NOx, CO_2, O_2, smoke).
- Orsat apparatus (chemical / absorption) : CO_2 (KOH), O_2 (alkaline pyrogallol), CO (cuprous chloride) absorbed in order; dry analysis.

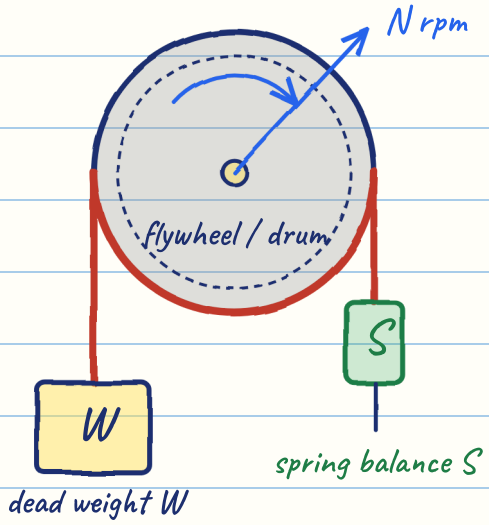
- **NDIR** (non-dispersive infrared) : CO, CO_2, HC. **FID** (flame ionisation) : total hydro-carbons. **Chemiluminescence (CLD)** : NO/NOx. **Paramagnetic** analyser / **zirconia lambda sensor** : O_2. **Gas chromatograph** : species.
- **Smoke meters** : Bosch (filter paper darkness, FSN), Hartridge (light extinction), opacimeter ; **PM** : gravimetric (dilution tunnel).
- **A/F from exhaust** : lean ⇒ high O_2; rich ⇒ high CO; stoichiometric ⇒ CO ≈ 0.

4.7 Measurement of Brake Power

- **Dynamometer** = device that **absorbs & measures** the engine power. Classified : **absorption** (all power converted to heat: rope brake, Prony brake, hydraulic, eddy-current, fan, electric (generating)) and **transmission** (power passes to a load : belt, epicyclic train, torsion dynamometers).

(a) Rope brake dynamometer

Rope brake dynamometer



Effective radius $R = (D + d)/2$

Net brake load = $(W - S)$

Torque $T = (W - S) \cdot R$ (N·m)

$BP = 2\pi NT / 60000$ (kW)

W, S in N (kg × 9.81)

D = drum dia, d = rope dia

Needs cooling water in the drum (heat = friction work)

- Simple & cheap, used for small engines. Drum is water-cooled. Rope turns around drum, with a dead weight on one end & spring balance on the other. **Disadvantage** : rope wears, load fluctuates, not for high power.

(b) Prony brake

- Two wooden **brake blocks** clamp over the flywheel through a bolt & spring; a **lever arm** (length l) carries a weight W (or load cell). $T = W \times l$, $BP = 2\pi NWl / 60000$. Adjusting the nuts changes the friction. Bad for high power as friction coefficient is unstable (chattering).

(c) Hydraulic (water-brake / Froude) dynamometer

- Rotor with pockets rotates in a casing filled with water; the **shearing & vortex flow** of water gives the resisting torque; torque arm measures reaction torque on the casing. Load regulated by water level/sluice. Handles large powers (up to many MW).

(d) Eddy-current dynamometer

- A steel/copper disc rotating between electromagnets (DC excited) generates **eddy currents** $\Rightarrow$ braking torque $\propto$ field current; the reaction measured by load cell. Quick response, fine control, widely used in modern test beds.

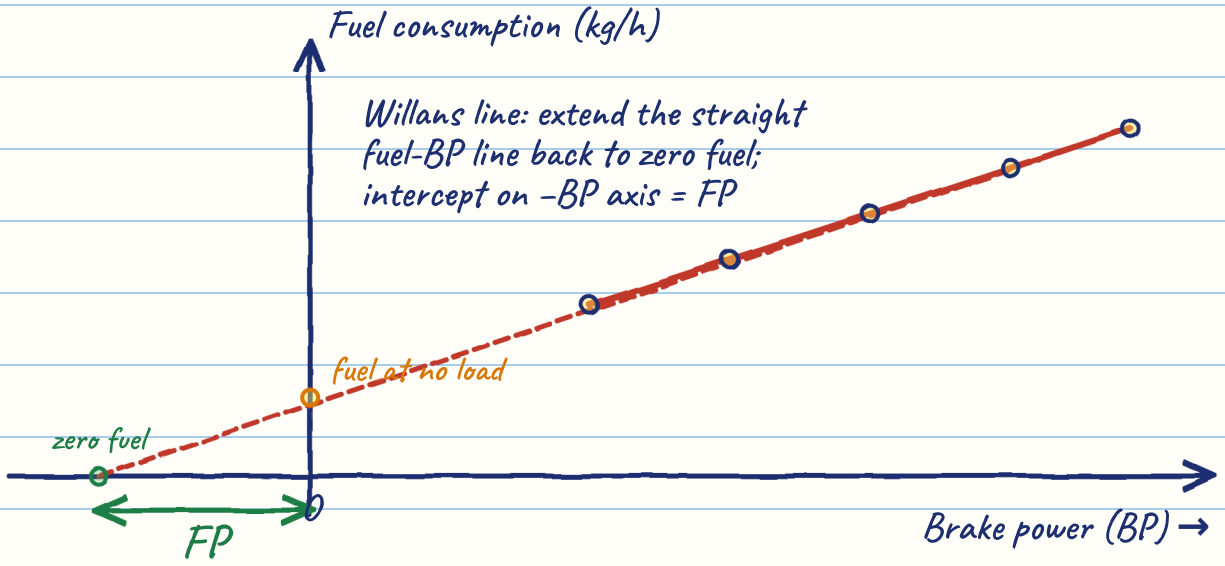
(e) Electric (swinging-field) dynamometer

- DC / AC generator with cradled stator; can both **absorb & drive (motoring)** the engine $\Rightarrow$ motoring test possible.

4.8 Determination of Frictional Power & Indicated Power

- **(1) Indicator diagram** - IP from imep (area / length × spring constant).

- (2) **Willans line method** (fuel-rack method) - used for **diesel engines** (smoke limit); at constant speed plot fuel consumption vs BP; extrapolate the (approximately straight) line back to **zero fuel** ⇒ intercept = −FP.



(3) Morse test (for multi-cylinder SI engines)

- The engine runs at constant speed at full throttle, BP measured. Then **one cylinder is cut out** (spark shorted) & the load reduced to bring the speed back to the same value; BP measured again. The reduction is the **IP of that cylinder.**

$$IP_1 = B - B_1,\ IP_2 = B - B_2,\ \ldots;\ IP = \Sigma IP_i;\ FP = IP - B$$

- Assumption : FP is the same with all cylinders firing or not.

Example (4-cyl) : B = 30 kW; B_1 = 20, B_2 = 19.5, B_3 = 20.5, B_4 = 19 kW ⇒ IP = 10 + 10.5 + 9.5 + 11 = **41 kW**; FP = 41 − 30 = **11 kW**; η_m = 30/41 = **73.2 %**.

(4) Motoring test

- The engine (fuelled off, hot) is **driven by an electric dynamometer** at rated speed; power absorbed = FP. Gives pumping + friction (compression work included) for all type engines. Drawback: the temperatures, pressure, & lube oil conditions differ from firing.

(5) Retardation (deceleration) test

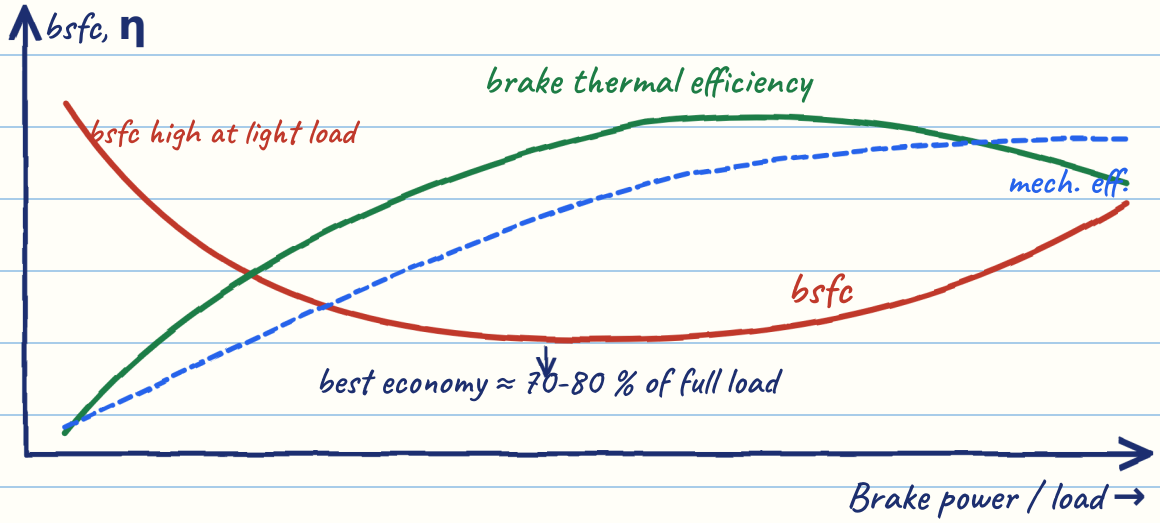
- Engine at rated speed, **cut-off** (ignition/fuel off) and the speed fall measured with time: with & without a load. FP ∝ moment of inertia × (dN/dt). Requires I (flywheel).

$$FP = (2\pi/60)^2 I N (dN/dt) / 1000$$

- **Comparison** : Morse test only for multi-cylinder SI ; Willans line for CI ; motoring for all (with electric dynamometer).

4.9 Performance Test

- Conducted to find BP, bsfc, η_m, η_{th}, η_v & draw performance curves. Types :
- (i) **Variable load test at constant speed** (e.g. generating set): load varies; measure BP, fuel, air, temperatures.
- (ii) **Variable speed at constant (full) load (throttle)** : find torque, power & bsfc vs speed - vehicle engines.
- (iii) **Morse test** & (iv) **Heat balance test** at rated speed.
- Observations : speed, load (weights, spring balance), fuel time, cooling-water flow & temperatures, exhaust temperature, air flow (manometer), ambient conditions.



- bsfc is **high at light loads** because of friction share; minimum at ≈ 70-80 % load; slightly increases near full load due to rich mixture. ηm increases with load (FP almost constant).

4.10 Heat Balance Sheet & Chart

- Distribution of the **heat supplied by the fuel** ($Q_s = m_f \times CV$) :

$$Q_s = BP + Q_w + Q_g + Q_{unacc.}$$

- (1) **Heat to BP** (kW), (2) **heat to cooling water** $Q_w = m_w c_w (T_{wo} - T_{wi})$, (3) **heat to exhaust gas** $Q_g = m_g c_{pg} (T_g - T_a)$ ($m_g = m_a + m_f$), (4) **unaccounted** (radiation, convection, friction, incomplete combustion, oil cooling).

Item	Typical SI	Typical CI
Brake power	25 - 30 %	30 - 38 %
Cooling water & oil	25 - 30 %	25 - 30 %
Exhaust gas	30 - 40 %	25 - 35 %
Unaccounted	5 - 10 %	5 - 8 %

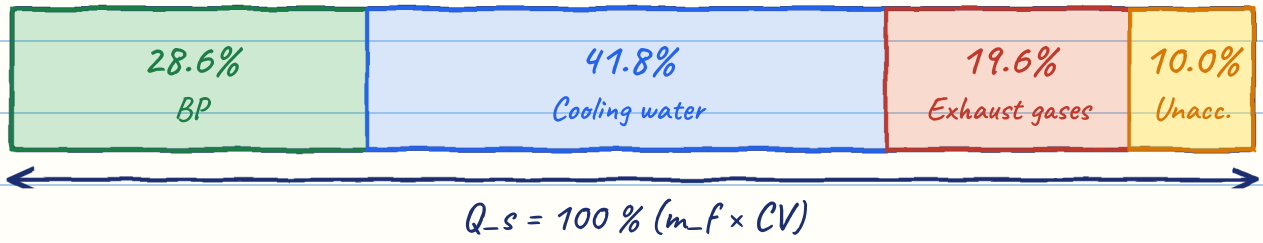
Heat balance sheet (kW and %) - Example

- 4-stroke diesel: BP = 8 kW, fuel = 2.4 kg/h (CV = 42000 kJ/kg), cooling water 0.08 kg/s with 35 K rise, air = 48 kg/h, exhaust temp 400 °C, ambient 27 °C, cp(gas) = 1.05 kJ/kg K.

- Heat supplied $Q_s = 2.4 \times 42000 / 3600 = 28$ kW (100 %).

Heat in	kW	%
Brake power	8.00	28.6
Cooling water (0.08 × 4.18 × 35)	11.70	41.8
Exhaust (50.4/3600 × 1.05 × 373)	5.48	19.6
Unaccounted (by difference)	2.82	10.0
Total	28.00	100

Heat balance chart (heat supplied by fuel = 100 %)



BP = useful work (28.6 %)
Cooling water = $m_w c_w (T_{out} - T_{in})$
Exhaust gas = $m_g c_{pg} (T_g - T_a)$
Unaccounted = Q_s − (BP + water + exhaust) → radiation, friction, etc.

4.11 Solved Problems

- **Problem 1** (BP by rope brake): drum $D = 0.6$ m, rope $d = 0.02$ m, $W = 40$ kg, $S = 5$ kg, $N = 600$ rpm.
  → $R = (0.6 + 0.02)/2 = 0.31$ m ; $(W - S) = 35 \times 9.81 = 343.35$ N ; $T = 343.35 \times 0.31 = 106.4$ N·m
  → $BP = 2\pi \times 600 \times 106.4 / 60000 = 6.69$ kW

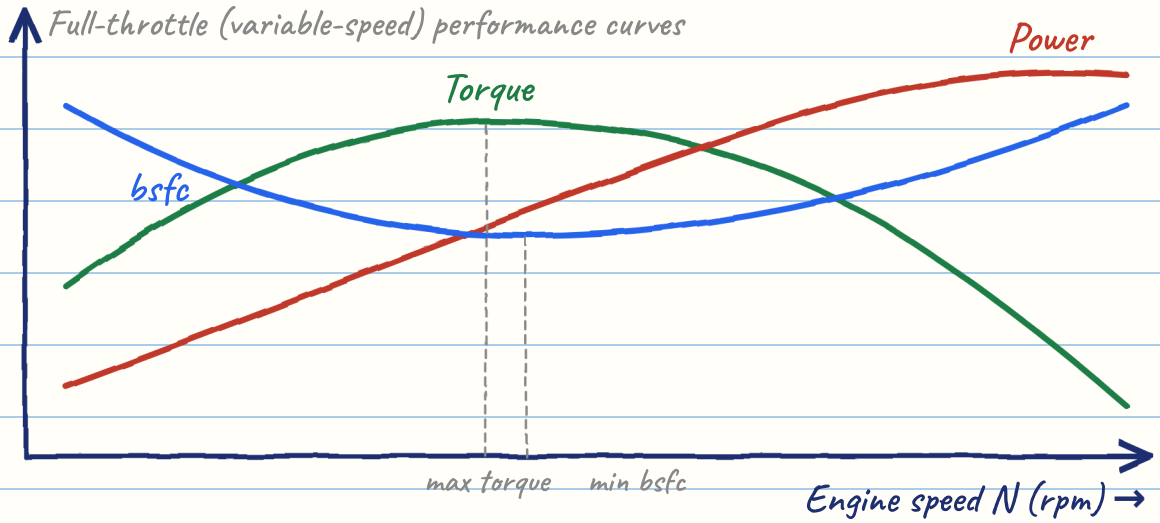
- **Problem 2** (IP from imep): 4-stroke, D = 100 mm, L = 150 mm, N = 1200 rpm, imep = 6 bar, 1 cylinder.
  → $A = (\pi/4)(0.1)^2 = 7.854\times10^{-3}\ m^2$; $IP = 6\times10^5 \times 0.15 \times 7.854\times10^{-3} \times (1200/2) / 60000 =$ **7.07 kW**
- **Problem 3** : the engine of Problem 2 has BP = 5.6 kW, uses 2.2 kg/h fuel of CV = 44000 kJ/kg.
  → $\eta_m = 5.6/7.07 =$ **79.2 %**; bsfc = 2.2/5.6 = **0.393 kg/kWh**
  → $\eta_{bth} = 5.6/(2.2 \times 44000/3600) = 5.6/26.9 =$ **20.8 %**; $\eta_{ith} = 7.07/26.9 =$ **26.3 %**
- **Problem 4** : bmep of a 4-cyl engine: BP = 30 kW, D = 90 mm, L = 100 mm, N = 3000 rpm.
  → $A = 6.362\times10^{-3}\ m^2$; $N_c = 3000/2 = 1500$; $bmep = 30 \times 60000 / (0.1 \times 6.362\times10^{-3} \times 4 \times 1500) = 4.72\times10^5\ N/m^2 =$ **4.72 bar**

> Remember : 1 bar = $10^5\ N/m^2$ = 100 kPa ; BP formula uses N in rpm ; $N_c = N/2$ (4-S).

4.12 Other Measurements in a Test Rig

- **Speed** : mechanical / digital **tachometer**, stroboscope, magnetic or optical pick-up (pulse counter), 60-tooth wheel.
- **Temperature** : thermocouples (K-type exhaust, J-type) , RTD (Pt-100) for water & oil; **pressure** : Bourdon gauge (oil), manometers (air).
- **Cooling-water flow** : rotameter / volumetric tank & stopwatch; **oil** : by sump gauge.
- **Calorific value of fuel** : bomb calorimeter (solid / liquid) , Junker's gas calorimeter (gas).

4.13 Variable-speed Performance Curves



- **Torque** rises with speed to a maximum (≈ 50-60 % of rated rpm, best filling) and then falls (volumetric efficiency & mechanical efficiency fall).
- **Power** increases almost linearly then flattens near rated speed (rises until torque falls faster than N rises); beyond rated speed power falls.
- **bsfc** is a "bathtub" curve, minimum at moderate speed (near max torque).
- Engine **flexibility** = ratio of speeds at max power to max torque; a flat torque curve = better vehicle driveability.

4.14 Formula Sheet

Quantity	Formula
IP	$p_{im} L A n (N/2) / 60000$ kW
BP	$2\pi NT/60000 = (W-S)\pi D_e N/60000$ kW
FP	IP − BP
η_m	BP / IP
bmep	BP × 60000 / (L A n N_c)
bsfc	m_f (kg/h) / BP (kW)

Quantity	Formula
η_{bth}	$BP / (\dot{m}_f \cdot CV)$, or $3600/(bsfc \cdot CV)$
η_v	V_{actual} / V_{swept}
Morse test	$IP_i = B - B_i$
Heat supplied	$\dot{m}_f \times CV$

4.15 Important Questions

- Define IP, BP, FP, bmep, bsfc, mechanical & thermal efficiency.
- Explain the Morse test & Willans-line method for finding the friction power.
- Describe the measurement of brake power using a rope brake / hydraulic dynamometer.
- Explain how air consumption & fuel consumption are measured in the laboratory.
- Prepare a heat-balance sheet; explain the chart & its significance.